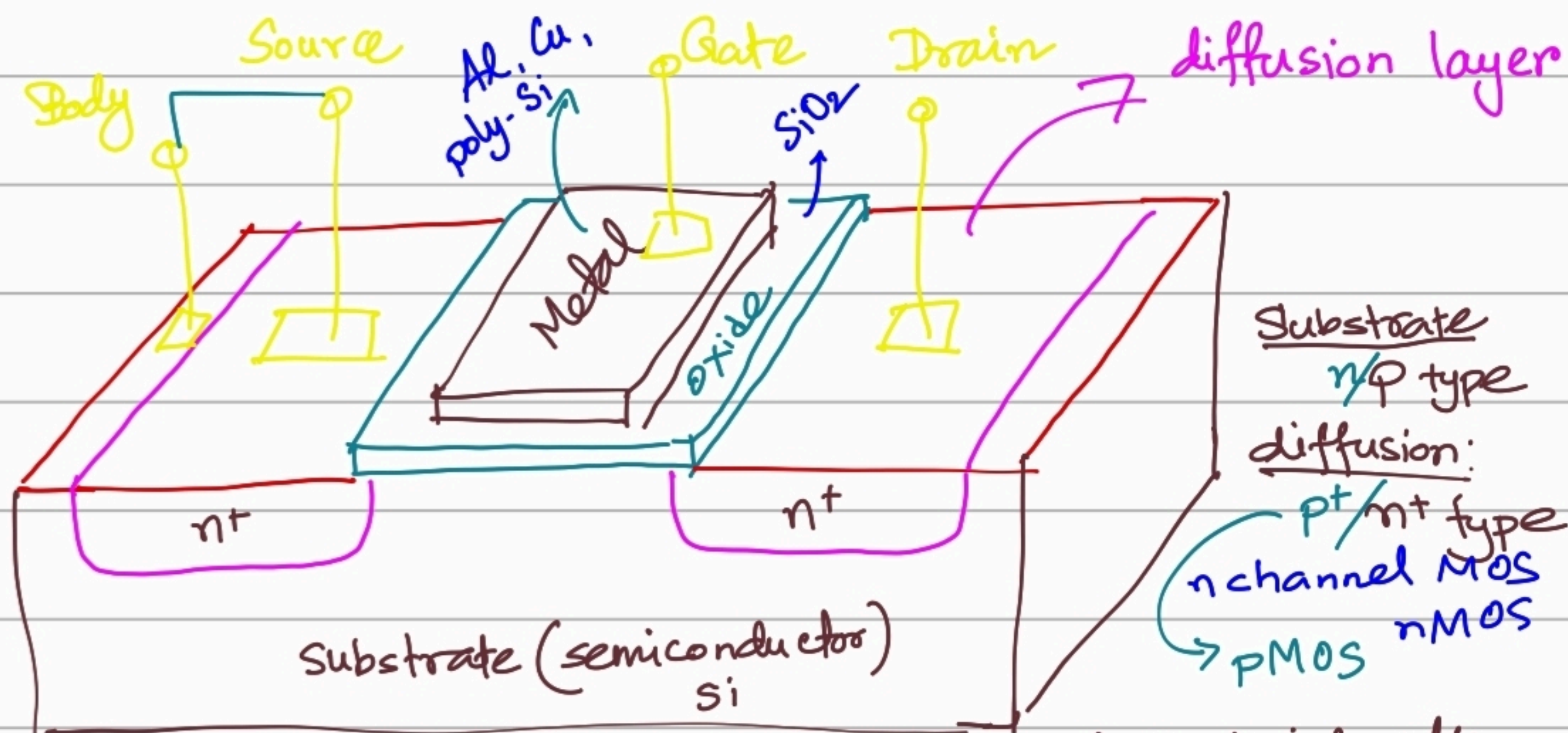


Lecture 12: MOS Logic family

Unipolar logic family: Current conduction is done by one type of charge carriers.

e.g. electron or hole.

Basic Internal Structure:

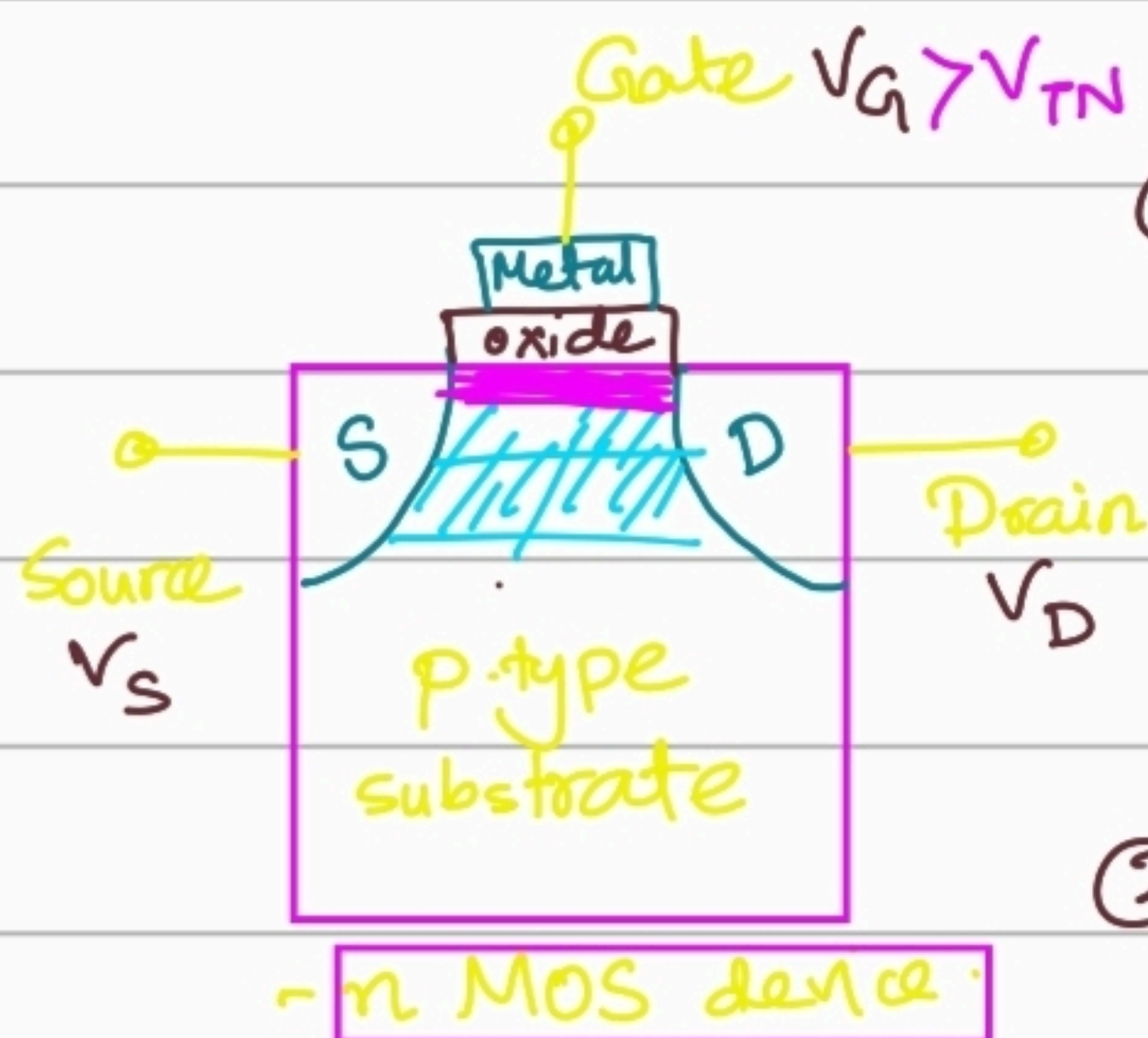


If some MOSFET is fabricated individually then source and body might be shorted.

In VLSI design we connect min/max for voltage is connected to body terminal, p-type/ ntype substrate.

Basic Operation:

If we apply voltage to the gate terminal.



① If V_G is less than zero, holes inside p-type semiconductor might accumulate underneath the oxide layer.

→ accumulation

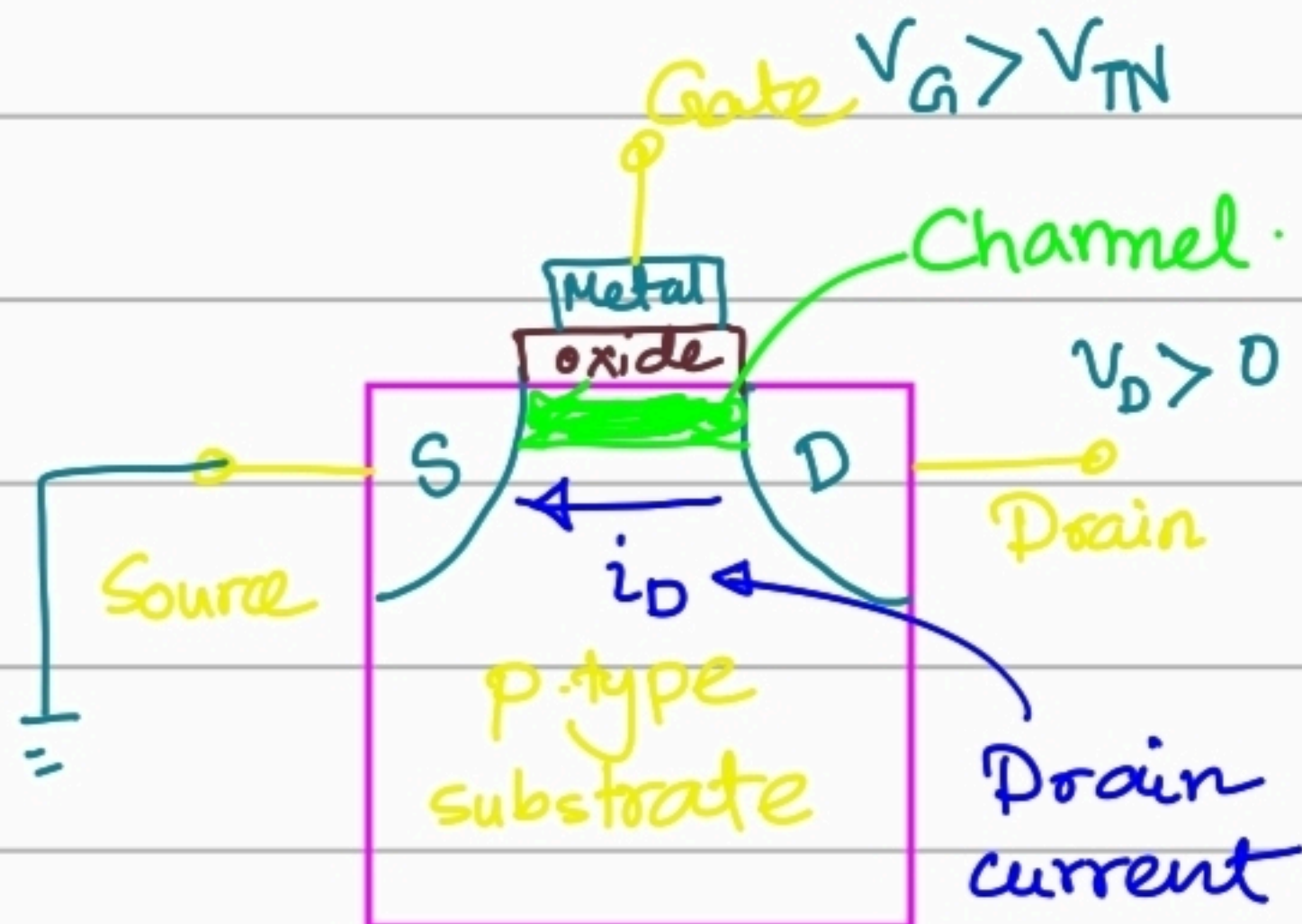
② If $0 < V_G < V_{TN}$, then depletion layer forms beneath the oxide layer.

→ depletion

③ If $V_G > V_{TN}$, then an inversion layer would emerge under the oxide layer. → Inversion

Now if we apply positive voltage to drain terminal when $V_G > V_{TN}$

① When drain voltage is very small positive voltage.

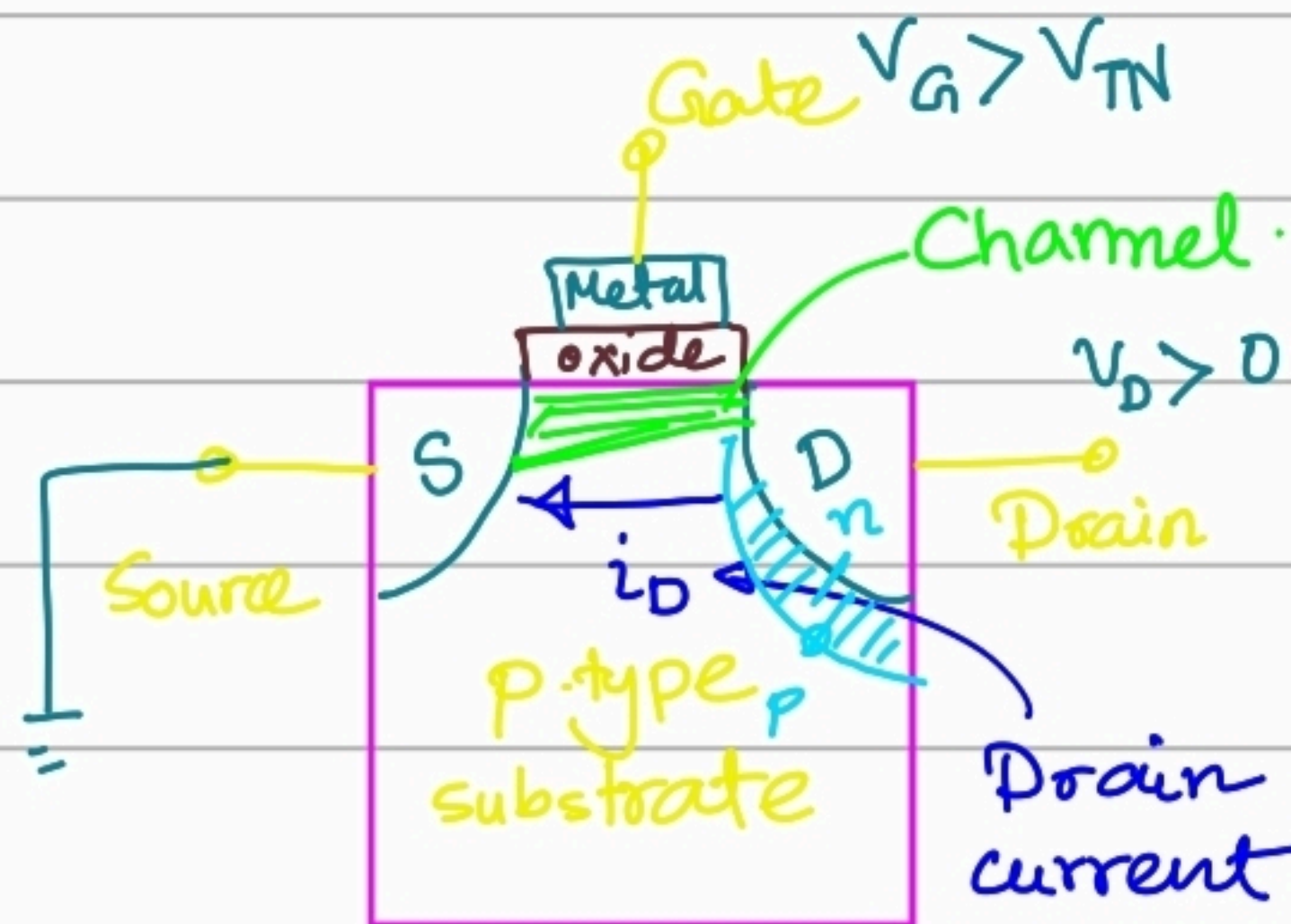


- n MOS device.

It behaves like a resistor.

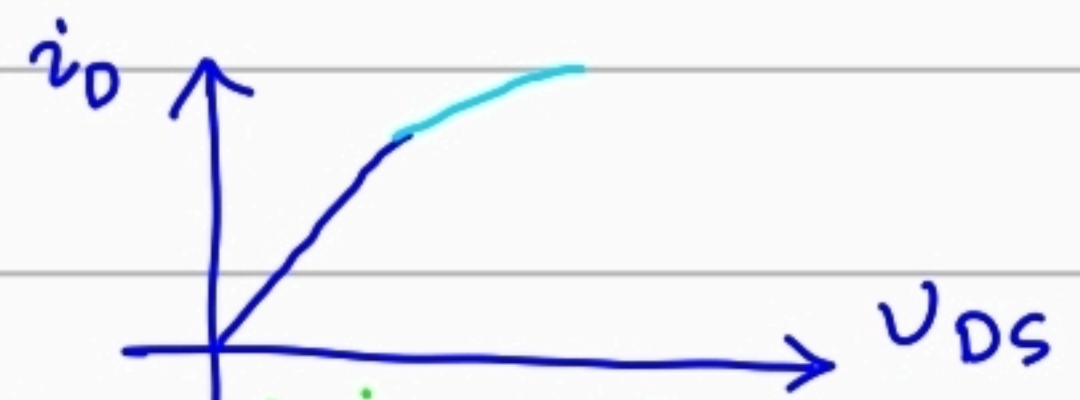


② When drain voltage is not very small.



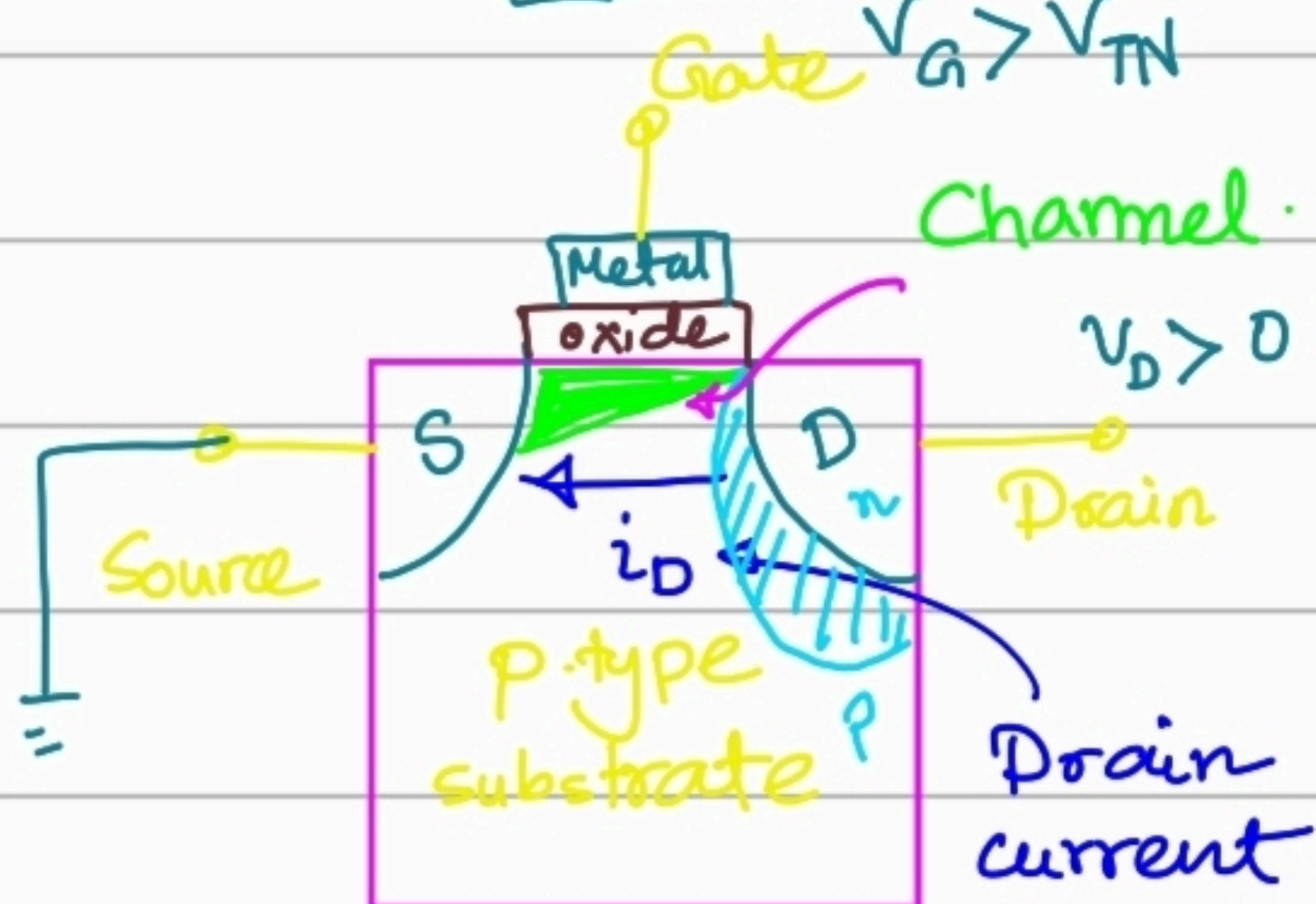
- n MOS device.

Drain-body p-n junction is in reverse bias



Channel is now shrinking near drain region

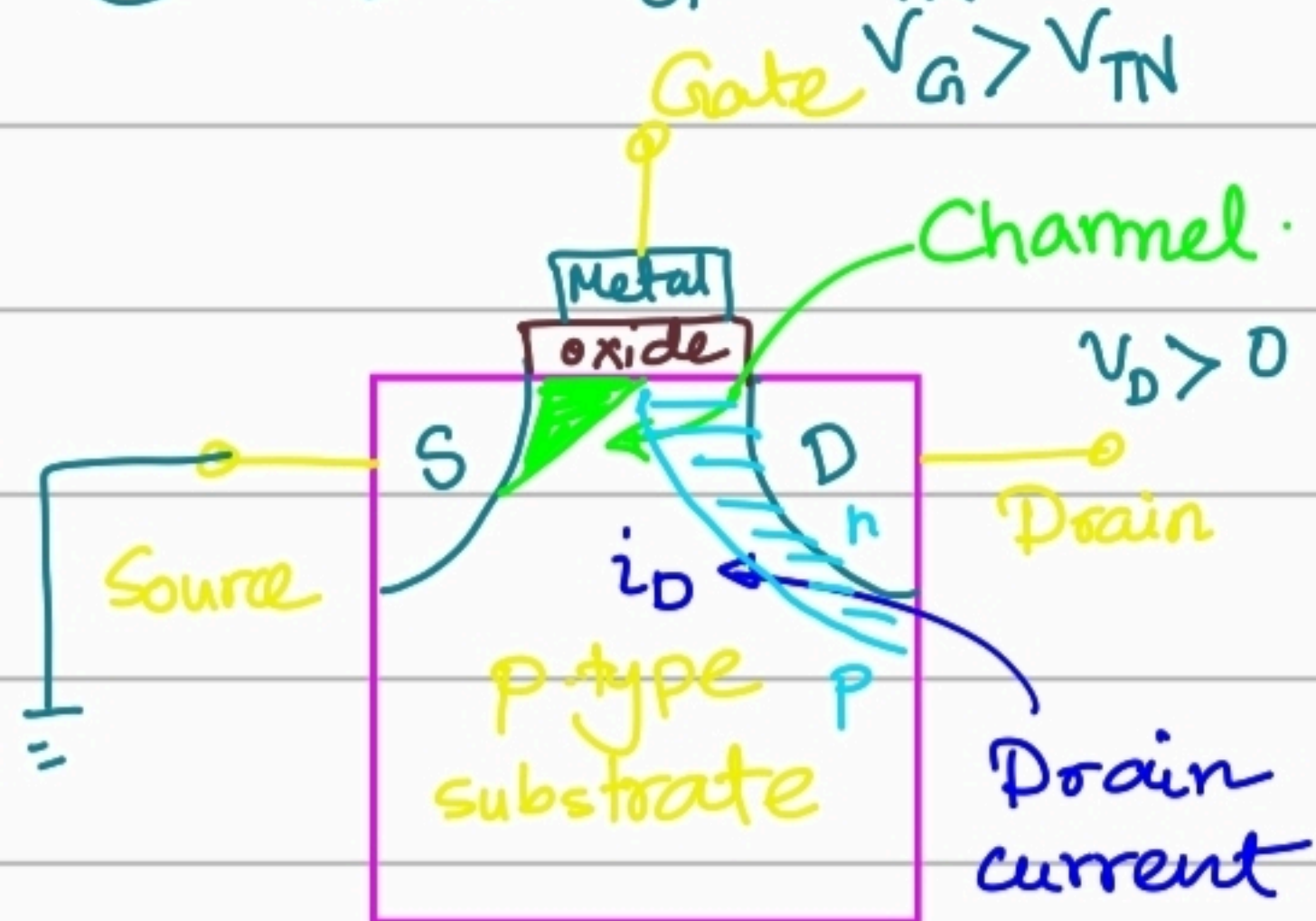
③ If the $V_D = V_G - V_{TN}$



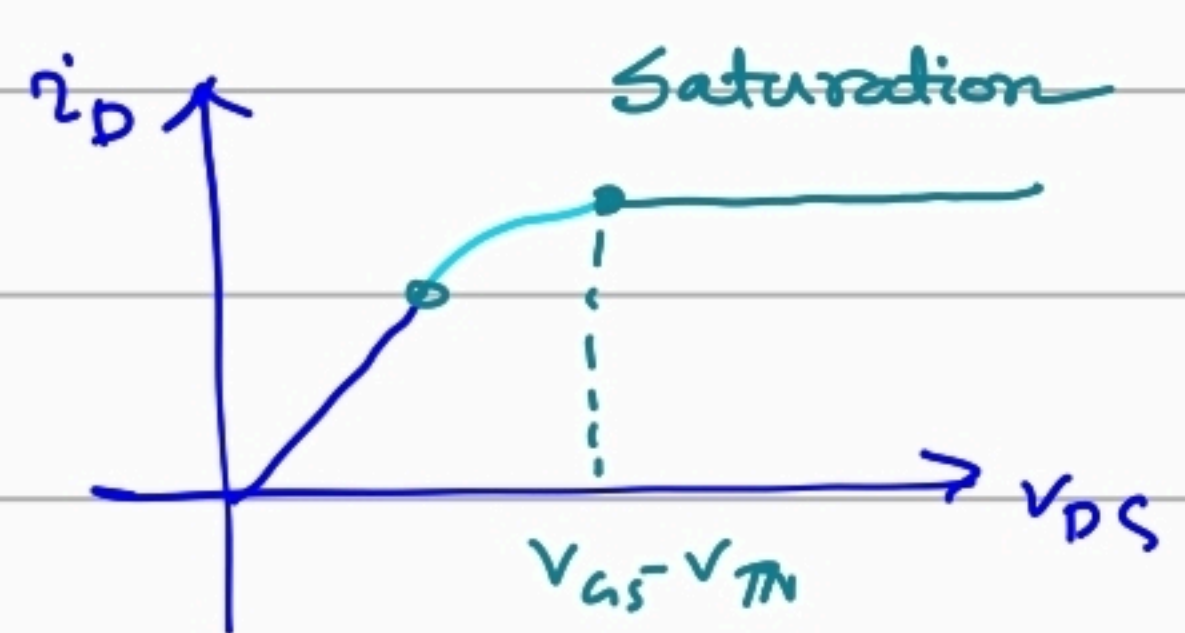
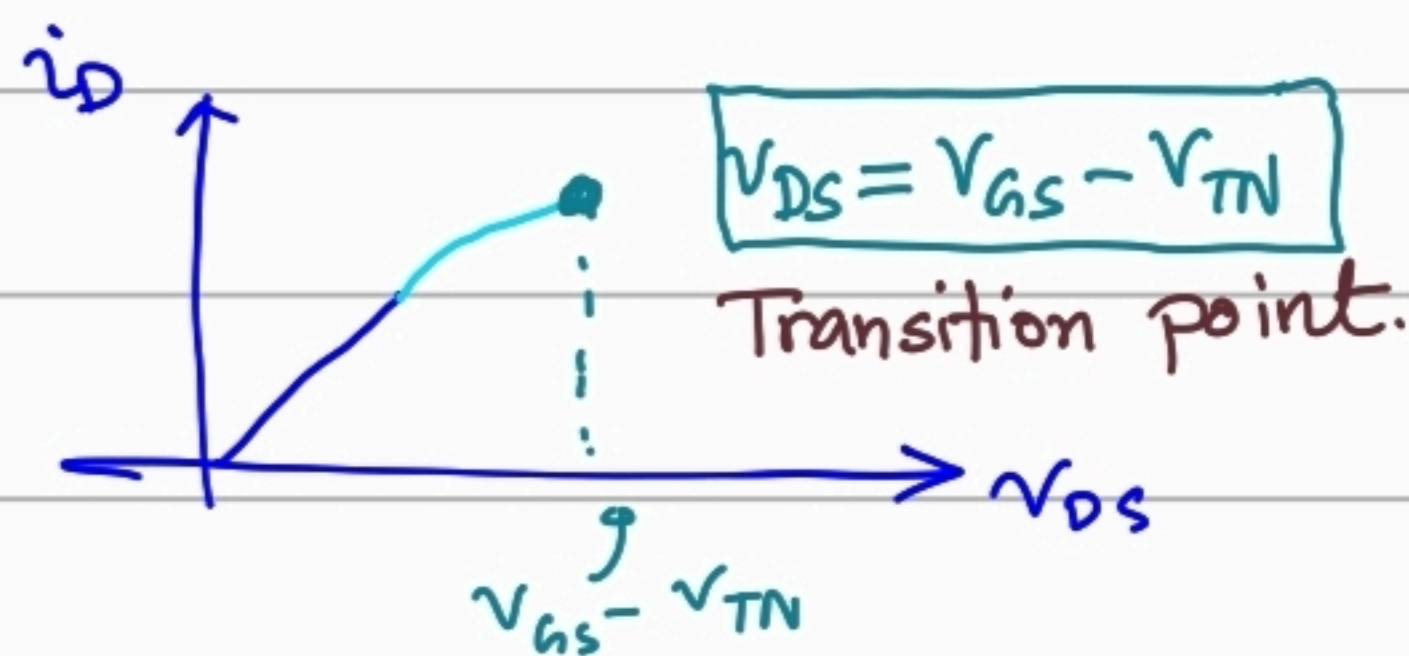
- n MOS device.

Channel region is now pinched off. Thus current cannot increase if we apply more voltage.

④ $V_D \geq V_G - V_{TN}$



- n MOS device.



Operating modes of MOS-FET

metal-oxide-semiconductor field-effect transistor

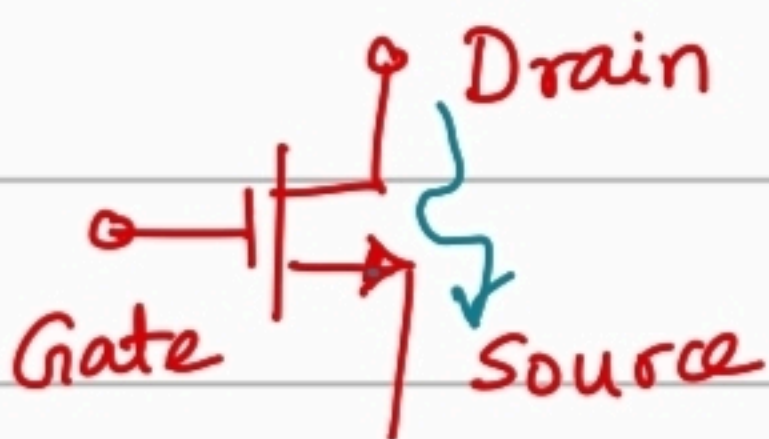
- ① Cutoff
- ② Linear/Triode
- ③ Saturation

We apply electric field to perform switching operation inside the transistor.

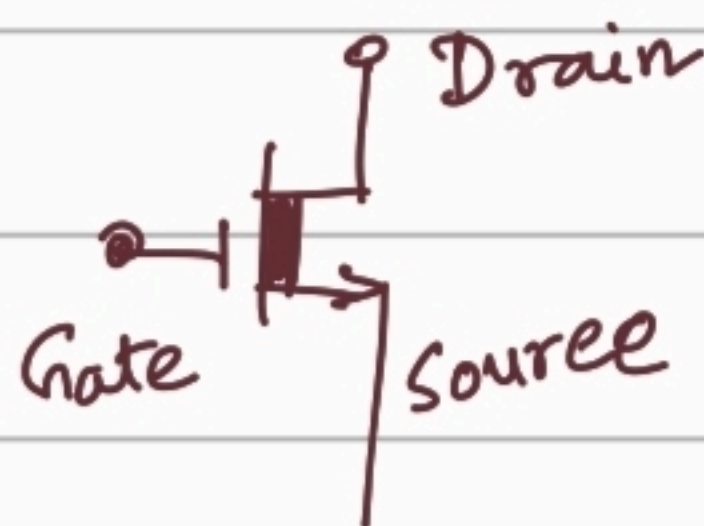
MOSFET types:

MOSFET type	Specialty	Cutoff	Triode	Saturation
n-channel enhancement	p-type sub. $V_{TN} > 0$	$V_{GS} < V_{TN}$	$V_{GS} > V_{TN}$ $V_{DS} \leq V_{GS} - V_{TN}$	$V_{GS} > V_{TN}$ $V_{DS} \geq V_{GS} - V_{TN}$
n-channel depletion	p-type sub. $V_{TN} < 0$	$V_{GS} < V_{TN}$	$V_{GS} > V_{TN}$ $V_{DS} \leq V_{GS} - V_{TN}$	$V_{GS} > V_{TN}$ $V_{DS} \geq V_{GS} - V_{TN}$
p-channel enhancement	n-type sub. $V_{TP} < 0$	$V_{SG} < V_{TP} $	$V_{SG} > V_{TP} $ $V_{SD} \leq V_{SG} + V_{TP}$	$V_{SG} > V_{TP} $ $V_{SD} \geq V_{SG} + V_{TP}$
p-channel depletion	n-type sub. $V_{TP} > 0$	$V_{SG} < V_{TP} $	$V_{SG} > V_{TP} $ $V_{SD} \leq V_{SG} + V_{TP}$	$V_{SG} > V_{TP} $ $V_{SD} \geq V_{SG} + V_{TP}$

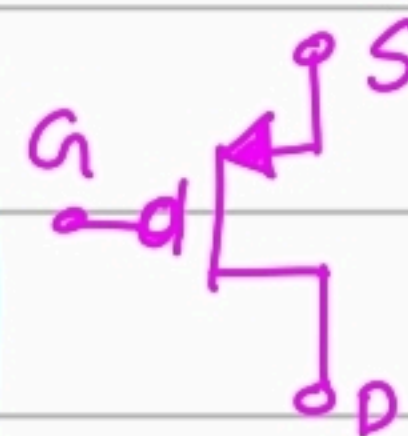
n channel
enhancement
MOSFET



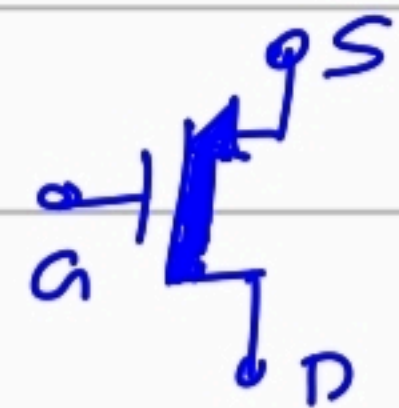
n channel
depletion
MOSFET



p-channel
enhancement
MOSFET



p-channel
depletion
MOSFET



Current-Voltage Relationships:

NMOS

① Cutoff mode: $i_D = 0$

② Triode mode:

$$i_D = K_n (2(V_{GS} - V_{TN})v_{DS} - v_{DS}^2)$$

③ Saturation mode:

$$v_{DS} = V_{GS} - V_{TN}$$

$$i_D = K_n (V_{GS} - V_{TN})^2$$

PMOS

① Cutoff mode: $i_D = 0$

② Triode mode:

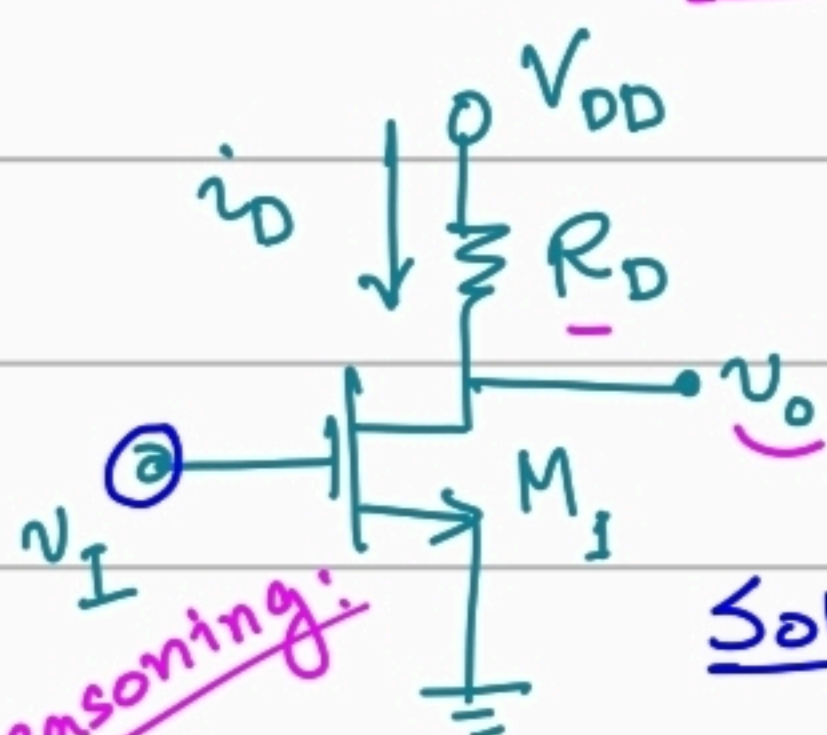
$$i_D = K_p (2(V_{SG} + V_{TP})v_{SD} - v_{SD}^2)$$

③ Saturation mode:

$$v_{SD} = V_{SG} + V_{TP}$$

$$i_D = K_p (V_{SG} + V_{TP})^2$$

Example: NMOS Inverter:



Q1: Determine v_O for $v_I = 5V$ and $v_I = 1.3V$.

Specification: $V_{DD} = 5V$, $R_D = 20K\Omega$.

$V_{TN} = 0.5V$, $K_n = 0.3mA/V^2$

Solⁿ: (a) $v_I = 5V$. $V_G = v_I = 5V$, $V_S = 0V$.

Reasoning:

Since this circuit will act like an inverter the higher the input voltage, the smaller the output voltage. $V_{GS} - V_{TN} = 5 - 0.5 = 4.5V > 0$. Thus $v_O < V_{GS} - V_{TN}$

So the M_1 might operate in triode region.

Assumption: $M_1 \rightarrow$ triode region.

$$i_D = K_n [2(V_{GS} - V_{TN})v_{DS} - v_{DS}^2] = \frac{V_{DD} - v_O}{R_D}$$

$$\Rightarrow 0.3m [2(5 - 0.5)v_O - v_O^2] = \frac{5 - v_O}{20K}$$

$$\Rightarrow \underbrace{20K \times 0.3m}_6 [9v_O - v_O^2] = 5 - v_O \Rightarrow 6v_O^2 - 55v_O + 5 = 0$$

$$v_O = \begin{cases} 9.074V \\ 0.092V \end{cases} \rightarrow i_D = \frac{5 - v_O}{20K} < 0 \text{ which is impossible.}$$

Thus. $v_O = 0.092V$

verification:

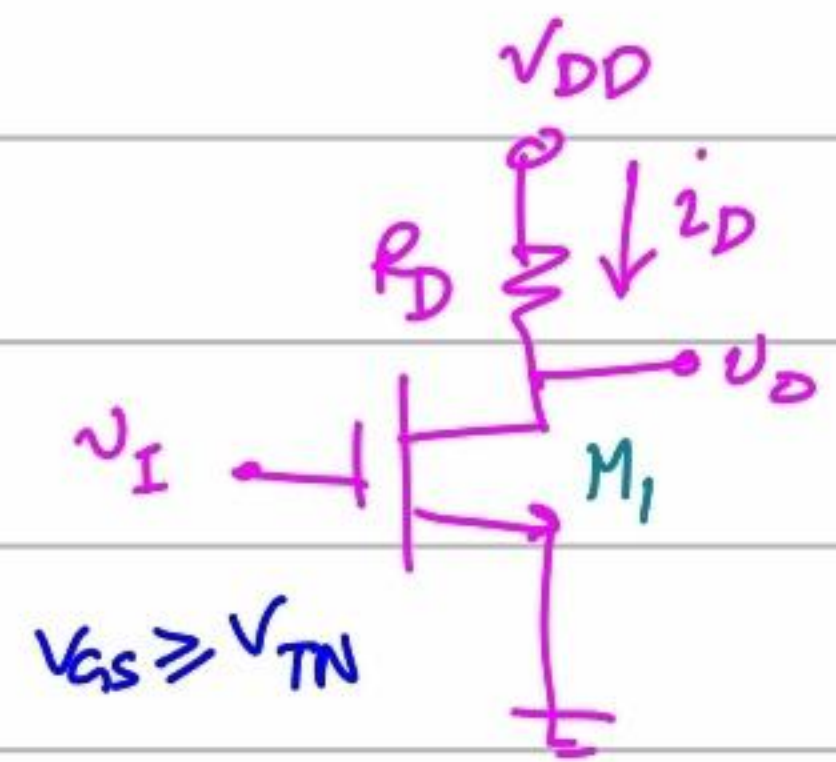
$$v_O = v_{DS} = 0.092 < 4.5 = (V_{GS} - V_{TN})$$

Therefore our assumption is valid.

⑥. $v_I = 1.5V$. Assumption: $M_1 \rightarrow$ saturation.

$$i_D = K_n [(V_{GS} - V_{TN})]^2$$
$$= 0.3 [1.5 - 0.5]^2 = 0.3mA$$

$$i_D = \frac{V_{DD} - v_o}{R_D} = \frac{5 - v_o}{20k} = 0.3mA$$
$$\Rightarrow v_o = -1V$$



$$v_{DS} = v_o = -1V < (V_{GS} - V_{TN}) = 1V$$

This contradicts our assumption that $v_{DS} \geq V_{GS} - V_{TN}$ (saturation)

Therefore, M_1 must be in triode/linear mode.

$$i_D = \frac{5 - v_o}{20} = 0.3 [2(1.5 - 0.5)v_o - v_o^2]$$

$$\Rightarrow 0.3v_o^2 + 0.65v_o - 0.25 = 0$$

$$v_o = \begin{cases} 1.6667V \\ \boxed{0.5V} \end{cases} \rightarrow v_{DS} > \begin{pmatrix} V_{GS} - V_{TN} \\ = 1V \end{pmatrix}$$

This cannot be the answer because M_1 must be in triode mode.

Therefore, $\boxed{v_o = 0.5V}$

Verification:

$$v_{DS} = 0.5 < (V_{GS} - V_{TN}) = 1V$$

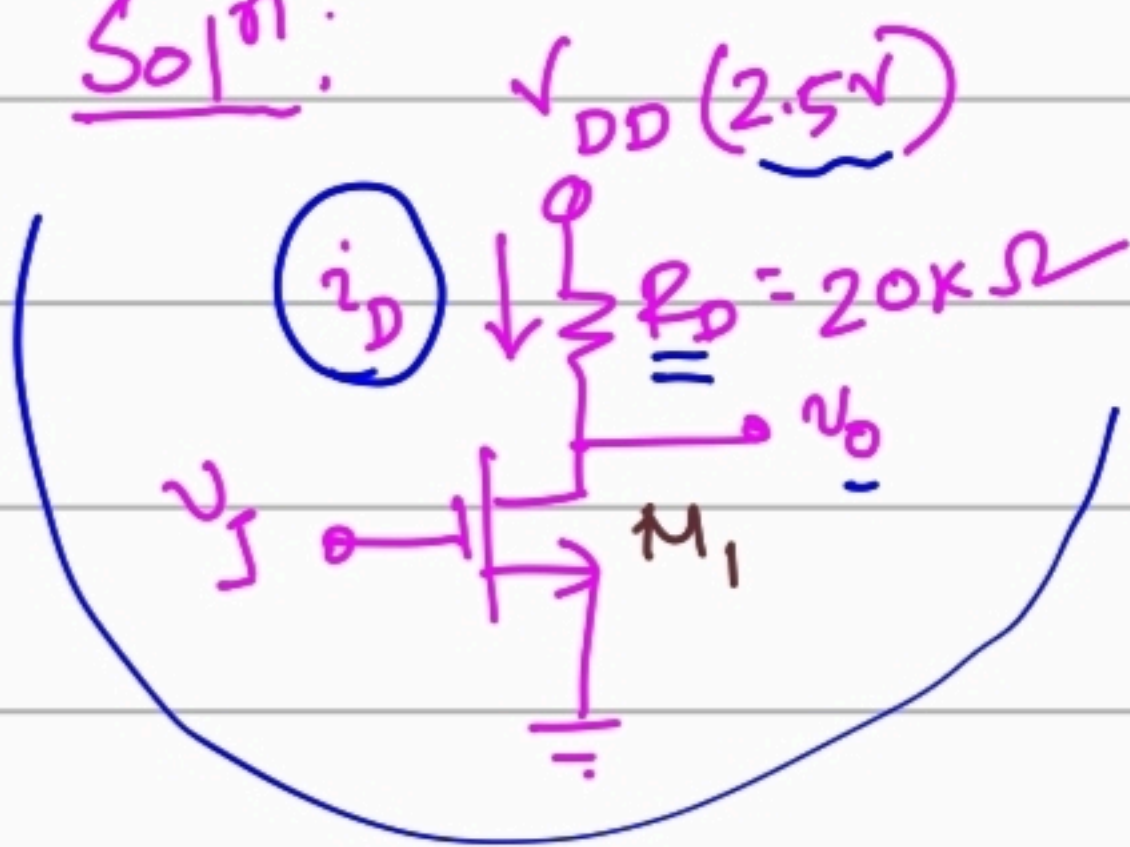
Thus our answer is not wrong.

Lecture 13: MOS logic family. CMOS logic-circuit

Example: Determine the transition point, maximum drain current and maximum power dissipation of an NMOS inverter with resistor.

Specification: $V_{DD} = 2.5V$, $R_D = 20K\Omega$, $V_{TN} = 0.5V$, $K_n = 0.3mA/V^2$

Solⁿ:



Observation: When output (v_O) has the smallest value then $i_D = \frac{V_{DD} - v_O}{R_D}$ would have a maximum value. Therefore, maximum input voltage is needed to produce this output value.

We can take maximum value of input voltage at 2.5V. $v_I = 2.5V$. [$M_1 \rightarrow$ triode]

$$i_D = \left(\frac{2.5 - v_O}{20K} \right) = 0.3m [2(2.5 - 0.5)v_O - v_O^2]$$

$$\Rightarrow 6v_O^2 - 25v_O + 2.5 = 0. \quad v_O = \begin{cases} 0.1025V \\ 4.064V \end{cases}$$

$v_O \neq 4.064V$ because that will produce negative i_D .
[$v_O = 0.1025V$]. [verify: $0.1025 < (V_{GS} - V_{TN}) = 2V$]

$$i_D = \frac{2.5 - 0.1025}{20K} = \boxed{0.1198mA}$$

$V_{DS} \leq (V_{GS} - V_{TN})$ $\uparrow V_{DD}$
 $\downarrow i_D$
max drain current

$$P = \Delta V I = (V_{DD} - 0) i_D = 2.5 \times 0.1198$$

maximum power dissipation = $\boxed{0.3mW}$

Transition point: [$V_{DS} = V_{GS} - V_{TN}$]

at transition point, $i_D = K_n (V_{GS} - V_{TN})^2$

$$i_D = \left(\frac{V_{DD} - v_O}{R_D} \right) = K_n (V_{GS} - V_{TN})^2 = K_n v_O^2 v_I$$

$$\Rightarrow \frac{2.5 - v_O}{20K} = 0.3m \cdot v_O^2 \Rightarrow 6v_O^2 + v_O - 2.5 = 0 \quad \checkmark$$



$$V_o = \begin{cases} 3.4051 \text{ V} \\ 0.5675 \text{ V} \end{cases} \rightarrow \text{if it were the sol}^n, i_D \text{ would be negative.}$$

$\therefore$ At transition point $V_o = 0.5675 \text{ V}$

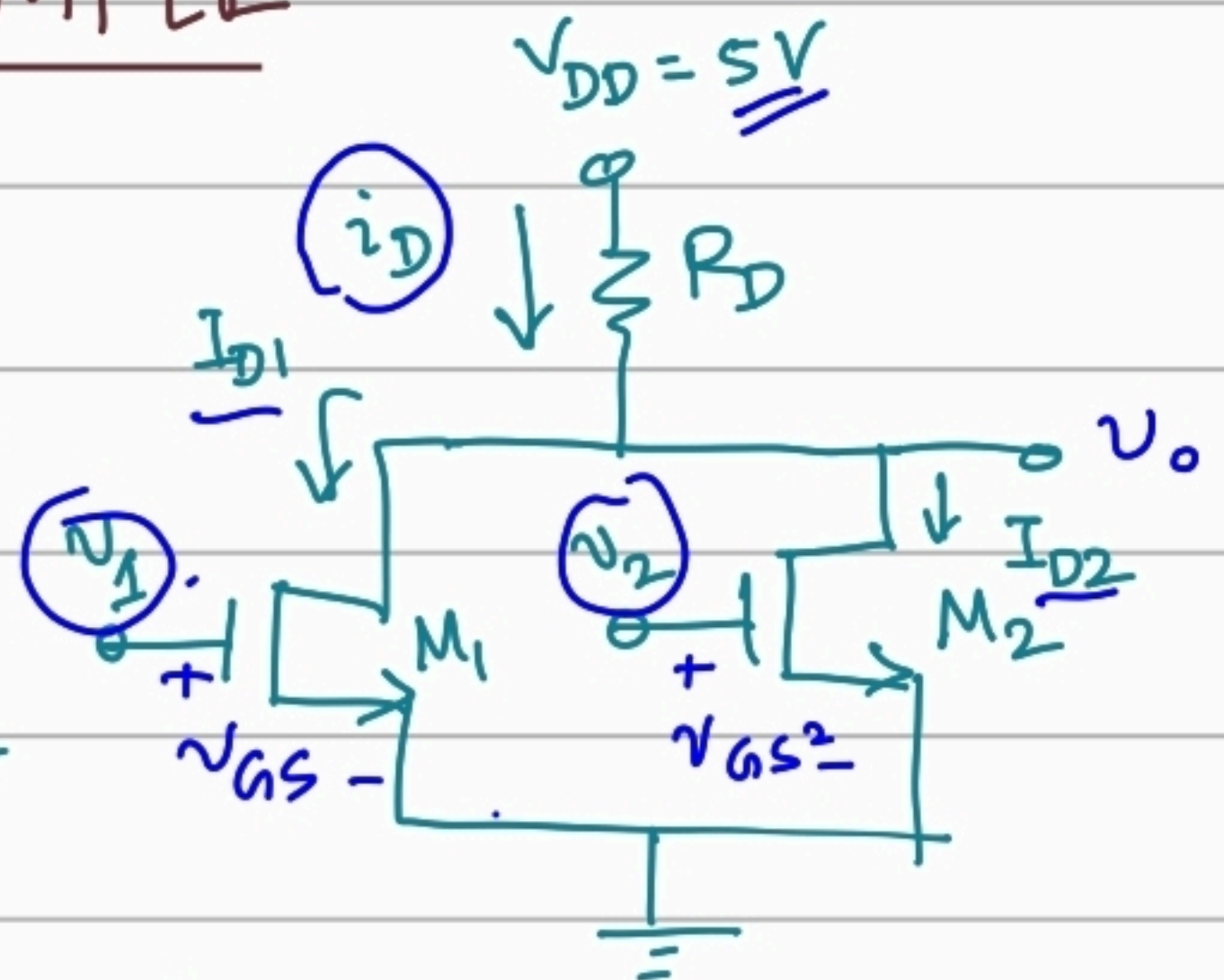
$$V_o = V_I - V_{TN}$$

$$V_I = V_o + V_{TN} = 1.0675 \text{ V}$$

NMOS NOR GATE EXAMPLE

Objective: Determine the current and voltages in this gate for different input conditions:

$$R_D = 20 \text{ k}\Omega, K_n = 0.1 \text{ mA/V}^2, V_{TN} = 0.8 \text{ V.}$$



Solⁿ: ① Case 1: $V_1 = V_2 = 0 \text{ V}$.

For M_1 we have $V_{GS1} = V_{G1} - V_{S1} = 0 - 0 = 0 < 0.8 \text{ V}$

" M_2 " " $V_{GS2} = V_{G2} - V_{S2} = 0 < 0.8 \text{ V}$

Therefore, M_1 and M_2 are in cutoff mode.

$$I_{D1} = I_{D2} = i_D = 0. \quad V_o = V_{DD} = 5 \text{ V.}$$

② Case 2: $V_1 = 5 \text{ V}$ and $V_2 = 0 \text{ V}$.

Since $V_{GS2} = 0 < 0.8$, M_2 must be in cutoff. $I_{D2} = 0$

So, we can assume M_1 is operating in triode region.

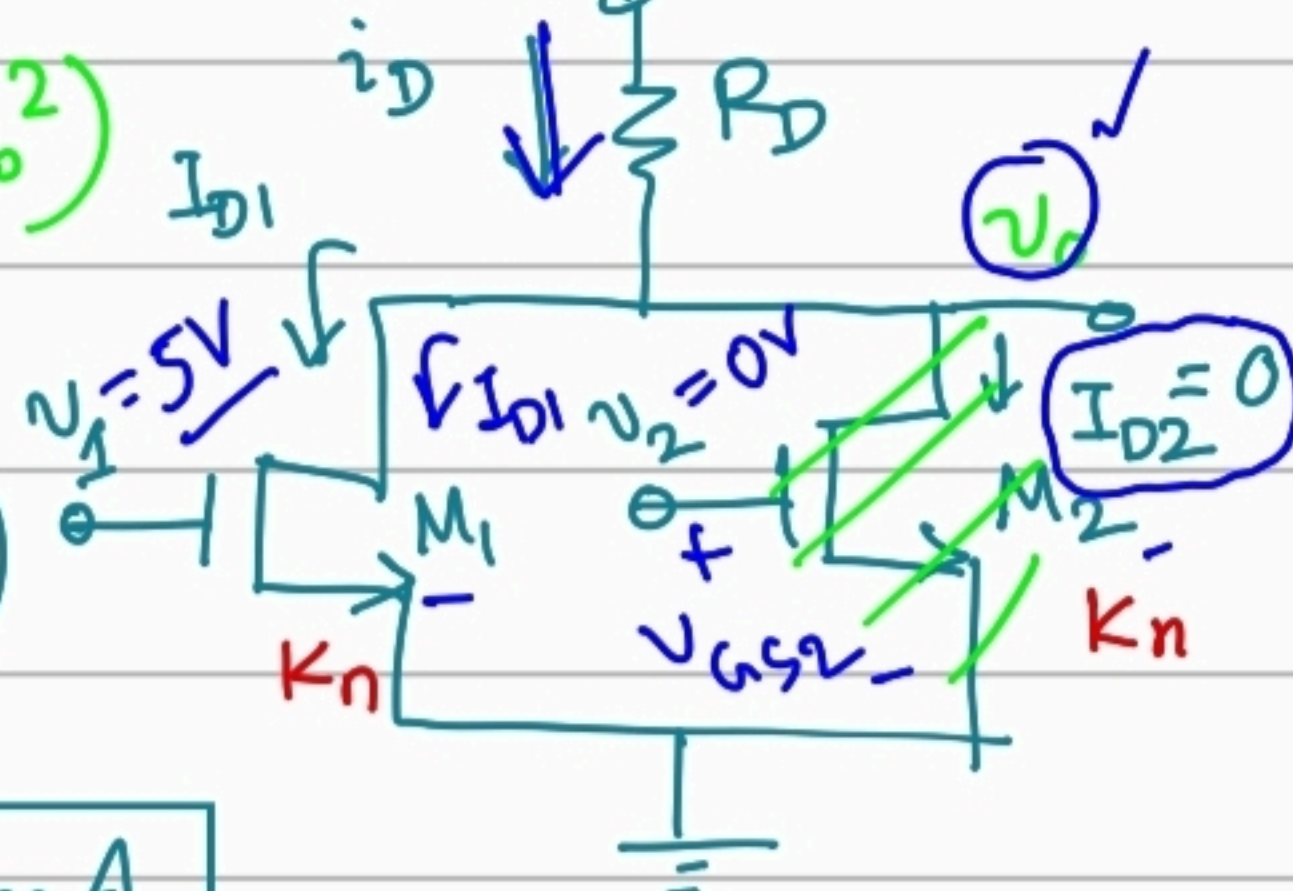
$$i_D = \frac{V_{DD} - V_o}{R_D} = K_n (2(V_{GS1} - V_{TN})V_{DS1} - V_{DS1}^2) \quad V_{DD} = 5 \text{ V}$$

$$\Rightarrow \left(\frac{5 - V_o}{20 \text{ k}} \right) = 0.1 \text{ m} (2(5 - 0.8)V_o - V_o^2)$$

$$\Rightarrow 2V_o^2 - 9.4V_o + 5 = 0 \quad \checkmark$$

$$V_o = \begin{cases} 8.6 \text{ V} \\ 0.29 \text{ V} \end{cases} \rightarrow (i_D < 0, \text{ impossible}) \quad \therefore V_o = 0.29 \text{ V}$$

$$i_D = I_{D1} = \frac{5 - 0.29}{20} = 0.2354 \text{ mA} \quad \checkmark$$



Case: ③. $V_1 = 0V$, $V_2 = 5V$. $M_1 \rightarrow \text{cutoff}$, $M_2 \rightarrow \text{Triode}$

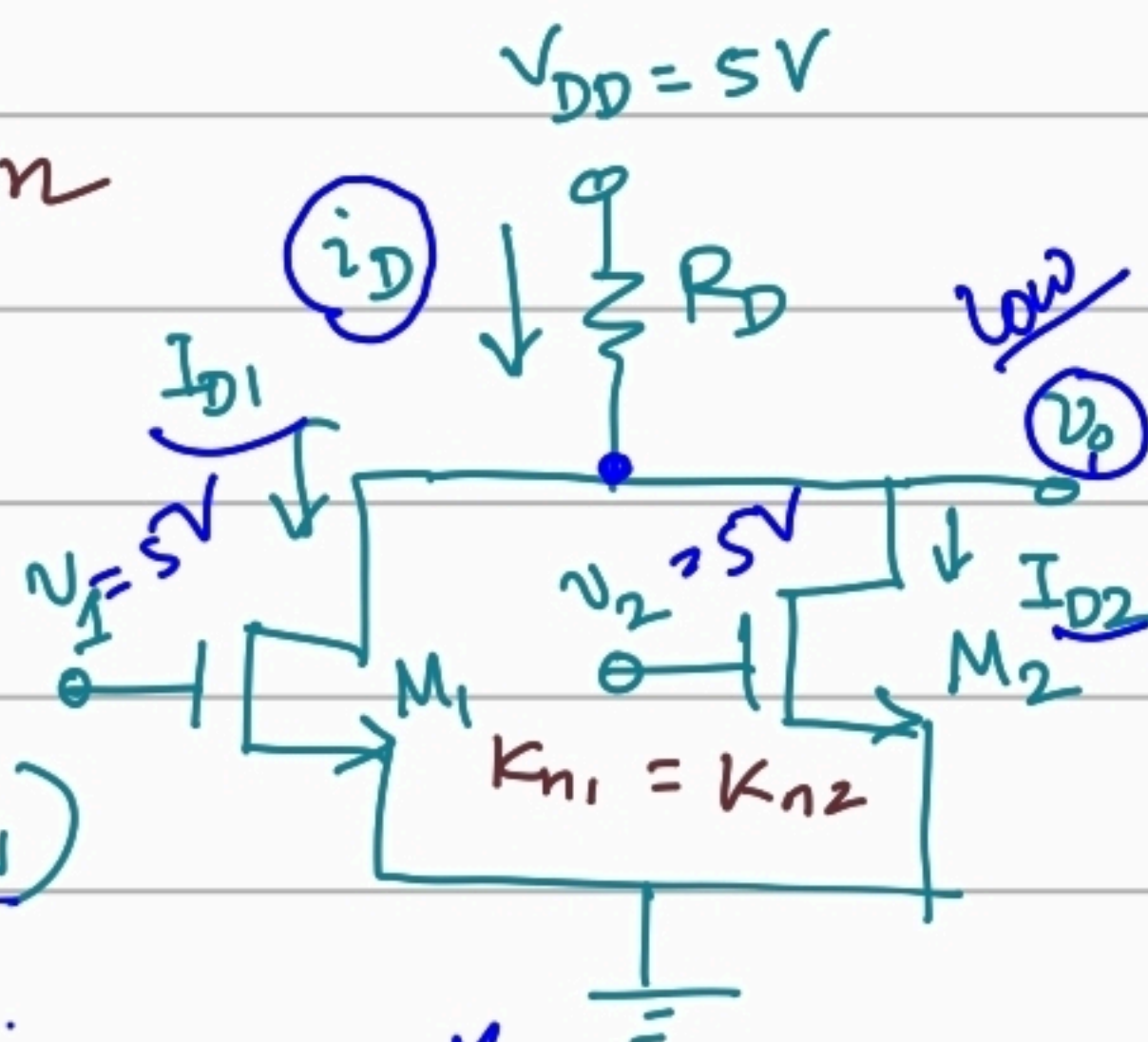
$v_o = 0.29V$, $i_D = 0.2354 \text{ mA}$ $[k_{n1} \neq k_{n2}]$

$I_{D1} = 0 \text{ mA}$, $I_{D2} = 0.2354 \text{ mA}$. $[V_{TN1} \neq V_{TN2}]$
 $[k_{n1} \neq k_{n2}]$

Case ④ $V_1 = V_2 = 5V$

Assuming, M_1 and M_2 are in triode region.

Reasoning: Because both inputs are high we might expect very low output voltage or drain voltage. ($V_{DS} < V_{GS} - V_{TN}$)



$i_D = I_{D1} + I_{D2} \leftarrow \text{using KCL.}$

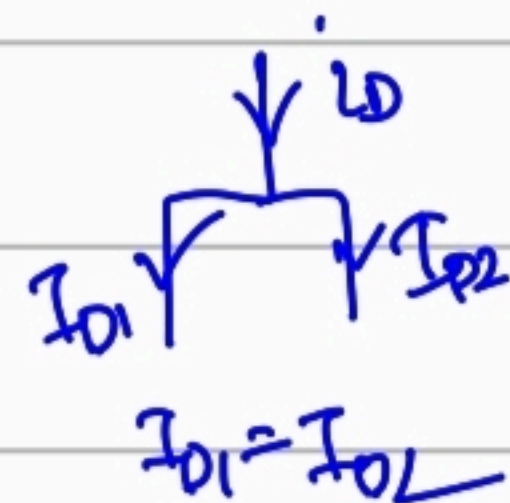
$\Rightarrow \frac{5 - v_o}{20K} = 0.1 \text{ m} [2 \times (5 - 0.8)v_o - v_o^2] \times 2$

$\Rightarrow 8v_o^2 - 34.6v_o + 5 = 0$. $v_o = \begin{cases} 4.17 \text{ V} \\ 0.149 \text{ V} \end{cases}$ $\rightarrow \text{Check yourself!}$

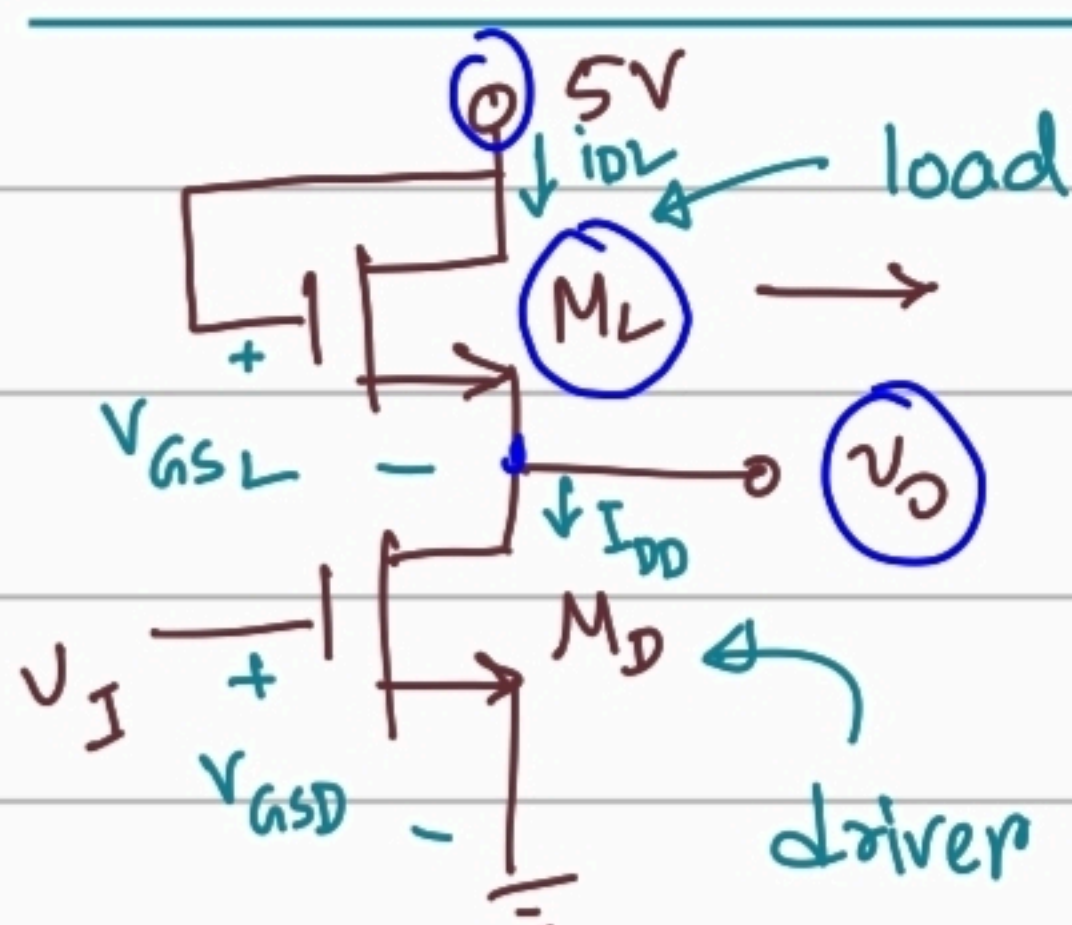
$v_o \neq 4.17 \text{ V}$ because, $v_{DS} = 4.17 \text{ V}$ which is much large value

$v_o = 0.149 \text{ V}$ $i_D = \frac{5 - 0.149}{20K} = 0.243 \text{ mA}$

$I_{D1} = I_{D2} = \frac{i_D}{2} = 0.121 \text{ mA}$



NMOS INVERTER with ENHANCEMENT LOAD



load nMOS is going to be enhancement type.

Determine the DC currents and voltages in the circuit containing enhancement load.

This is only true for MOSFET

Specifications: $V_{TND} = V_{TNL} = 1V$, $k_{nD} = 50 \mu\text{A/V}^2$

$k_{nL} = 10 \mu\text{A/V}^2$. Determine v_o , when $V_I = 5V$ and $V_I = 1.5V$

Observation regarding load nMOS:

For M_L , we can see that

$V_{G_L} = V_{D_L} = 5V$. Therefore,

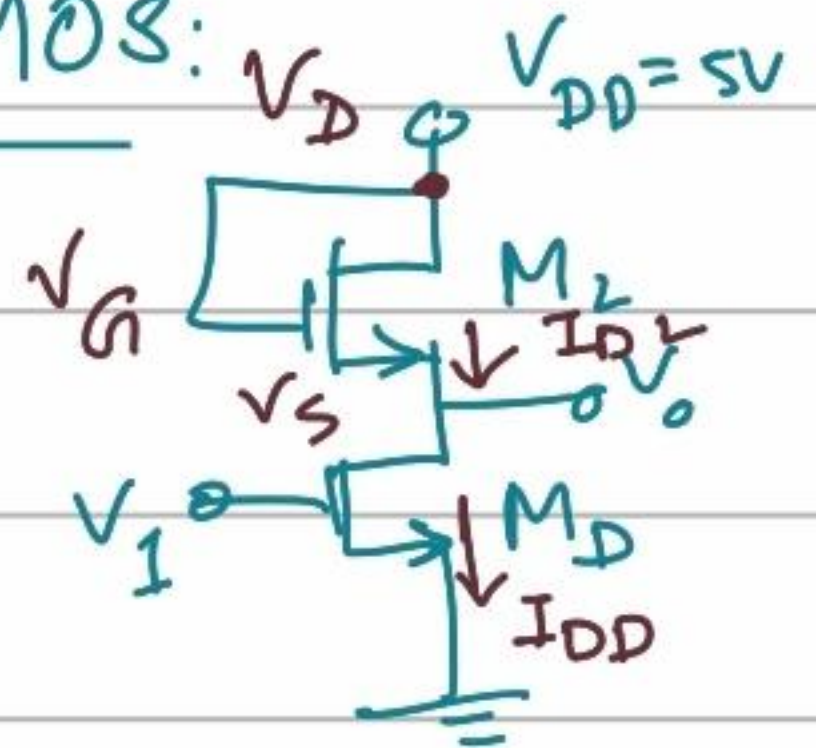
$$V_{D_S L} = V_{D_L} - V_{S_L} = V_{G_L} - V_{S_L}$$

$$\Rightarrow V_{D_L} - V_{S_L} \geq (V_{G_L} - V_{S_L}) - V_{TNL}$$

$$\Rightarrow V_{D_S L} \geq (V_{G_S L}) - V_{TNL}$$

Therefore M_L satisfies saturation condition regardless of input and output voltage.

Thus, M_L must be in saturation mode for all cases.



Now for M_D , we are assuming saturation mode. [we are expecting output voltage or $V_{D_S D}$ is high compared to $V_{G_S D} - V_{TND}$]

Using KCL, $I_{D_L} = I_{D_D}$

$$\Rightarrow K_{nL} (V_{G_S L} - V_{TNL})^2 = K_{nD} (V_{G_S D} - V_{TND})^2$$

$$\Rightarrow 0.01 (5 - V_0 - 1)^2 = 0.05 (1.5 - 0 - 1)^2$$

$$\Rightarrow V_0 = 2.88V. \Rightarrow V_{D_S D} = 2.88 > \underbrace{V_{G_S D} - V_{TND}}_{0.5}$$

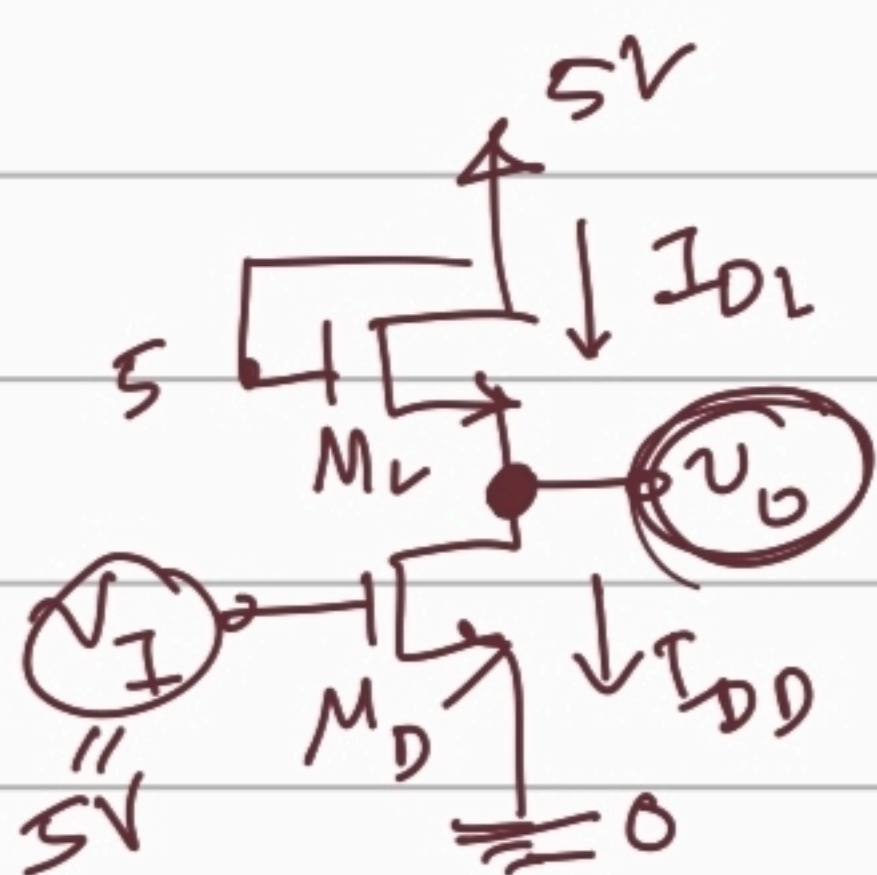
Our assumption is valid.

$$\therefore I_{D_L} = I_{D_D} = 0.05 (1.5 - 1)^2 = 12.5 \mu A$$

$$= 1.25 \times 10^{-2} mA$$

Case ②: $V_I = 5V$. [Because input is high we are expecting to get small output voltage, thus drain voltage is small therefore it might be less than $V_{GS} - V_{TN}$]

So we are assuming M_D is in triode mode



$$I_{DL} = I_{DD}$$

$$K_{nL}(V_{GSL} - V_{TNL})^2 = K_{nD}(2(V_{GSD} - V_{TND})(V_{DSD} - V_{DSD}^2))$$

$$\Rightarrow \frac{10}{1000}(5 - v_o - 1)^2 = \frac{50}{1000}(2(5 - 0 - 1)v_o - v_o^2)$$

$$\Rightarrow 6v_o^2 - 48v_o + 16 = 0$$

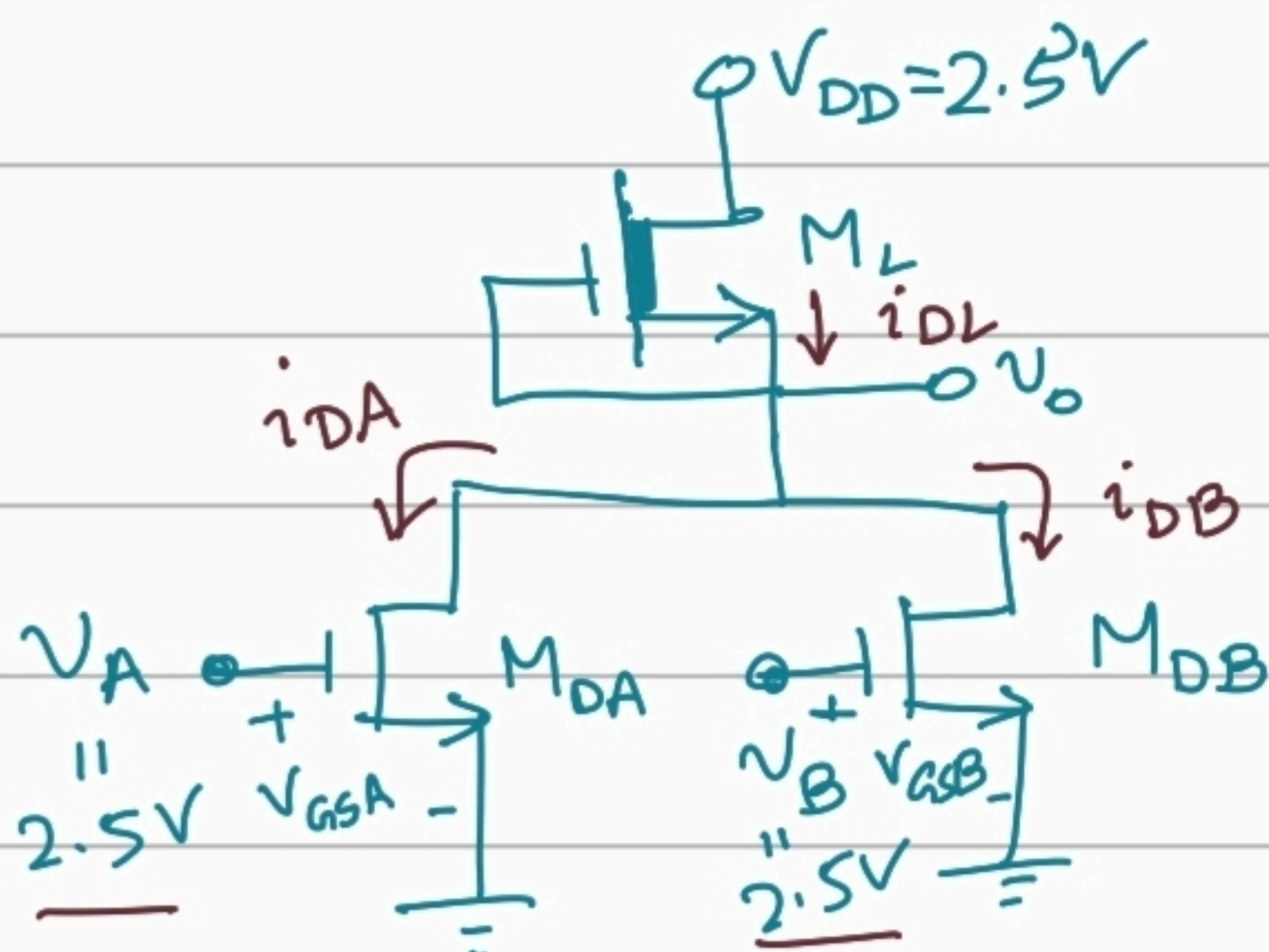
$$\Rightarrow v_o = \begin{cases} 7.65V \\ 0.34V \end{cases} \Rightarrow \text{Here } 7.65V > 5V = V_{DD} \text{ which is impossible}$$

Therefore, $v_o = 0.34V$. $I_D = \frac{10}{1000}m(5 - 0.34 - 1)^2$

$$I_{DL} = I_{DD} = 0.133mA$$

★

(Quiz 4) NMOS NOR GATE with Depletion Load



Q: Determine the output voltage of NMOS NOR gate.

$$K_n' = 100\mu A/V^2 = 0.1mA/V^2$$

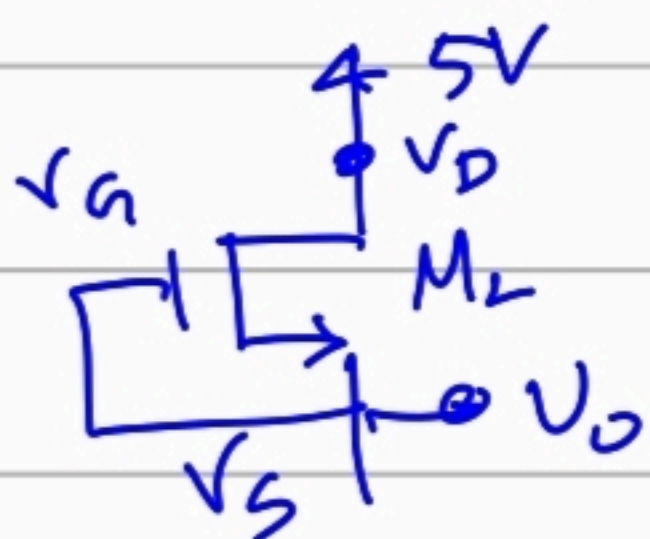
$$V_{TND} = V_{TNA} = V_{TNB} = 0.4V$$

$$V_{TNL} = -0.6V$$

$$\left(\frac{W}{L}\right)_D = 4, \left(\frac{W}{L}\right)_L = 1.$$

$$K_n = K_n' \left(\frac{W}{L}\right)$$

Solⁿ: Observation: As this is a NOR gate when two input voltages are high, output voltage is low. Therefore v_o is small.



For M_L depletion type NMOS we

have $V_{GSL} = V_{GL} - V_{SL} = 0V$

$V_{DSL} = V_{DL} - V_{SL} = 5 - v_o \rightarrow$ this value is large.

Thus. $V_{DSL} \geq V_{GSL} - V_{TNL} = 0 - (-0.6) = 0.6$

$5 - v_o \geq 0.6$ [Because we expect v_o to be small]

Hence, for this particular input condition we can deduce that M_L must be in saturation mode.

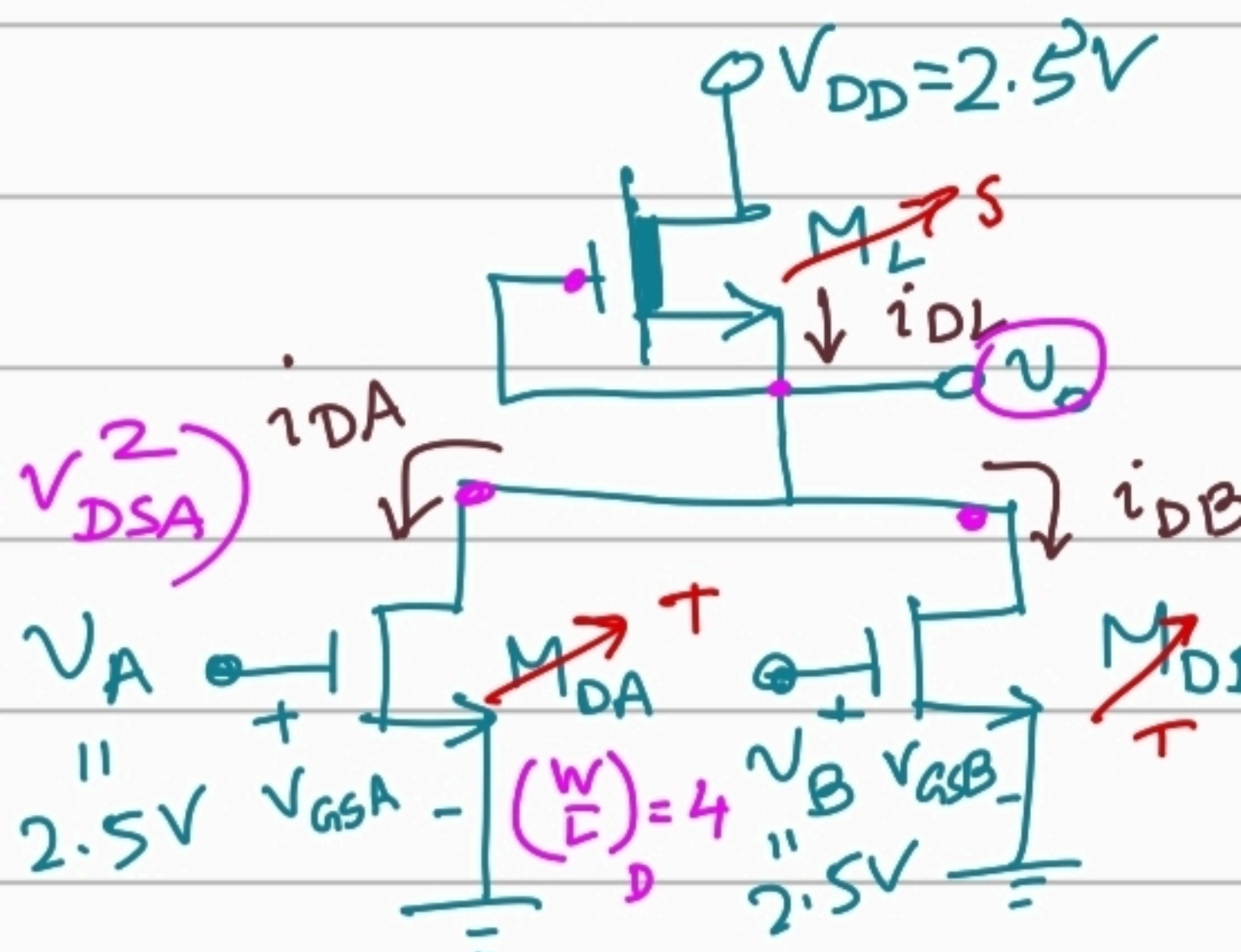
We are also expecting M_{DA} and M_{DB} to be in triode mode because v_o or drain voltage is smaller than gate voltage minus threshold voltage.

$i_{DL} = i_{DA} + i_{DB}$

$\Rightarrow K_{nL}(V_{GSL} - V_{TNL})^2$

$= K_{nA}(2(V_{GSA} - V_{TNA})V_{DSA} - V_{DSA}^2)$

$+ K_{nB}(2(V_{GSB} - V_{TNB})V_{DSB} - V_{DSB}^2)$



$\Rightarrow \cancel{K_n} \left(\frac{W}{L} \right)_L (0 - (-0.6))^2$

$= \cancel{K_n} \left(\frac{W}{L} \right)_D (2(2.5 - 0 - 0.4)v_o - v_o^2) + \cancel{K_n} \left(\frac{W}{L} \right)_D 2(2.5 - 0.4)v_o - v_o^2$

$$\Rightarrow 4v_o^2 - 16.8v_o + 0.36 = 0$$

$$\underline{v_o} = \begin{cases} 4.178V \\ 0.02153V \end{cases} \rightarrow v_o \geq 2.5 = v_{DD} \text{ which is not possible.}$$

Finally, the solution. $v_o = 0.02153V$.

$$i_{DL} = k'_n \left(\frac{W}{L}\right)_L (V_{GSL} - V_{TNL})^2$$

$$= \frac{100}{1000} \times 1 (0 - (-0.6))^2 = \boxed{0.036mA}$$

$$i_{DA} = k'_n \left(\frac{W}{L}\right)_D (2(V_{GSA} - V_{TNA})V_{DSA} - V_{DSA}^2)$$

$$= \frac{10}{1000} \times 4 (2(2.5 - 0.4)0.02153 - 0.02153^2)$$

$$= \boxed{0.018mA}$$

$$i_{DB} = k'_n \left(\frac{W}{L}\right)_D (2(V_{GSB} - V_{TNB})V_{DSB} - V_{DSB}^2)$$

$$= \boxed{0.018mA}$$